Homework #2 solution (total 100 pts.)

1 (Total 25 pts.) A system consisting of a uniform rigid bar of mass m and length L, and with both ends mounted on rollers, is constrained to move as shown in **Fig. 2.23**. The left and right ends of the bar are attached to linear springs of stiffnesses k_1 and k_2, respectively. When the bar is horizontal the springs are unstretched, write answers to the following questions.

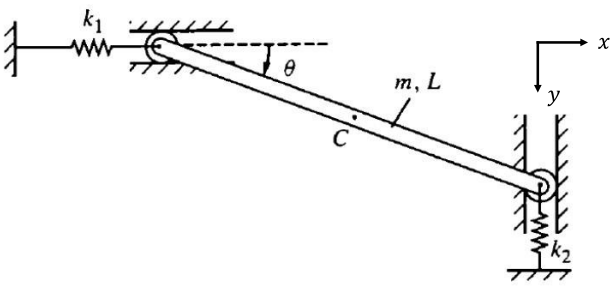


Figure 2.23 Rigid bar in constrained motion

1) **(20 pts)** Derive the equations of motion for the system as shown **Fig 2.23**, using the <u>Newtonian approach</u>.

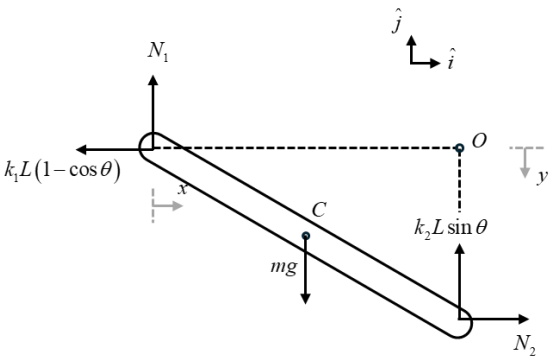
(Hint: Two equations for the translation of the mass center C and one equation for the rotation about C. The angle θ can be arbitrarily large.)

2) **(5 pts)** When the system is given with the following parameters:

$$m = 1 \text{ kg},\ L = 2 \text{ m},\ k_1 = 5 \text{ N/m},\ k_2 = 10 \text{ N/m}$$

Based on the result obtained in 1), derive the equations of motion for the system.

1. Free Body Diagram (FBD),



2. For equation of motion of C, (displacement/velocity/acceleration)

$$r_c = \left(-\frac{L}{2}\cos\theta\right)\hat{\imath} + \left(-\frac{L}{2}\sin\theta\right)\hat{\jmath}$$

$$\mathrm{v_c} = \left(+\frac{L}{2}\sin\theta\cdot\dot{\theta}\right)\hat{\imath} + \left(-\frac{L}{2}\cos\theta\cdot\dot{\theta}\right)\hat{\jmath}$$

$$\mathrm{a_c} = \left(\frac{L}{2}\cos\theta\cdot\dot{\theta}^2 + \frac{L}{2}\sin\theta\cdot\ddot{\theta}\right)\hat{\imath} + \left(+\frac{L}{2}\sin\theta\cdot\dot{\theta}^2 - \frac{L}{2}\cos\theta\cdot\ddot{\theta}\right)\hat{\jmath}$$

3. For equation of force, (x, y, & moment)

$$\sum F_x = m\left(\frac{L}{2}\cos\theta\cdot\dot{\theta}^2 + \frac{L}{2}\sin\theta\cdot\ddot{\theta}\right) = -k_1L(1-\cos\theta) + N_2$$

$$\sum F_y = m\left(\frac{L}{2}\sin\theta\cdot\dot{\theta}^2 - \frac{L}{2}\cos\theta\cdot\ddot{\theta}\right) = k_2L\sin\theta + N_1 - mg$$

$$\sum M_c = \frac{1}{12}mL^2\times\ddot{\theta} = -k_1L(1-\cos\theta)\cdot\frac{L}{2}\sin\theta + N_1\frac{L}{2}\cos\theta - \mathrm{k_2L}\sin\theta\cdot\frac{\mathrm{L}}{2}\cos\theta - \mathrm{N_2}\frac{\mathrm{L}}{2}\sin\theta$$

$$\downarrow$$

$$\boldsymbol{-k_1L(1-\cos\theta) + N_2 - m\left(\frac{L}{2}\cos\theta\cdot\dot{\theta}^2 + \frac{L}{2}\sin\theta\cdot\ddot{\theta}\right) = 0}$$

$$\boldsymbol{k_2L\sin\theta + N_1 - mg - m\left(\frac{L}{2}\sin\theta\cdot\dot{\theta}^2 - \frac{L}{2}\cos\theta\cdot\ddot{\theta}\right) = 0}$$

$$\boldsymbol{-k_1L(1-\cos\theta)\cdot\frac{L}{2}\sin\theta + N_1\frac{L}{2}\cos\theta - \frac{1}{12}mL^2\times\ddot{\theta} - k_2L\sin\theta\cdot\frac{L}{2}\cos\theta - N_2\frac{L}{2}\sin\theta = 0}$$

4. Based on the equations of motion and given parameters:

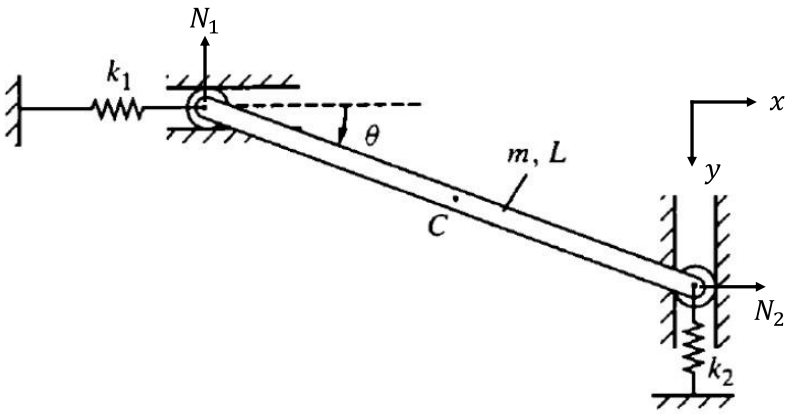


Figure 2.23.1 Rigid bar in constrained motion

$$\boldsymbol{-k_1L(1-\cos\theta) + N_2 - m\left(\frac{L}{2}\cos\theta\cdot\dot{\theta}^2 + \frac{L}{2}\sin\theta\cdot\ddot{\theta}\right) = 0}$$

$$\boldsymbol{k_2L\sin\theta + N_1 - mg - m\left(\frac{L}{2}\sin\theta\cdot\dot{\theta}^2 - \frac{L}{2}\cos\theta\cdot\ddot{\theta}\right) = 0}$$

$$\boldsymbol{-k_1L(1-\cos\theta)\cdot\frac{L}{2}\sin\theta + N_1\frac{L}{2}\cos\theta - \frac{1}{12}mL^2\times\ddot{\theta} - k_2L\sin\theta\cdot\frac{L}{2}\cos\theta - N_2\frac{L}{2}\sin\theta = 0}$$

$$\downarrow\ m = 1\text{ kg},\ L = 2\text{ m},\ k_1 = 5\text{ N/m},\ k_2 = 10\text{ N/m},\ N_1 = 3\text{ N},\ N_2 = 6\text{ N},\ g = 10\ m/s^2$$

$$-10(1-\cos\theta) + 6 - \left(\cos\theta\cdot\dot{\theta}^2 + \sin\theta\cdot\ddot{\theta}\right) = 0$$

$$20\sin\theta - 7 - \sin\theta\cdot\dot{\theta}^2 + \cos\theta\cdot\ddot{\theta} = 0$$

$$-10(1-\cos\theta)\sin\theta + 3\cos\theta - \frac{1}{3}\ddot{\theta} - 20\sin\theta\cos\theta - 6\sin\theta = 0$$

2 (Total 25 pts.) Using Hamilton's principle, derive the equation of motion for the system shown in **Fig. 2.25**, by answering the following questions step by step.

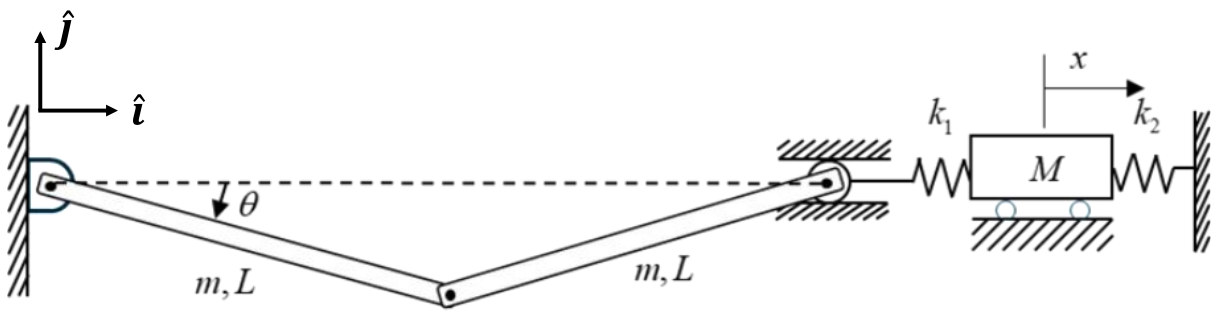


Figure 2.25 System consisting of two links and a mass

1) **(5 pts)** The total kinetic energy T of the system
2) **(5 pts)** The total potential energy V of the system.
3) **(5 pts)** Formulate the Lagrangian of the system. (Hint: $L = T - V$)
4) **(10 pts)** Using the Lagrangian obtained in 3), derive the equation of motion of the system by means of Hamilton's principle.

1. Equations of motion,

$$r_A = \frac{L}{2}\cos\,\theta\hat{\imath} - \frac{L}{2}\sin\,\theta\hat{\jmath} \qquad v_A = -\frac{L}{2}\sin\,\theta\cdot\dot{\theta}\hat{\imath} - \frac{L}{2}\cos\,\theta\cdot\dot{\theta}\hat{\jmath}$$
$$r_B = \frac{3}{2}L\cos\,\theta\hat{\imath} - \frac{L}{2}\sin\,\theta\hat{\jmath} \qquad v_B = -\frac{3}{2}L\sin\,\theta\cdot\dot{\theta}\hat{\imath} - \frac{L}{2}\cos\,\theta\cdot\dot{\theta}\hat{\jmath}$$
$$r_C = x\hat{\imath} \qquad v_C = \dot{x}\hat{\imath}$$

2. Lagrangian,

$$T = \frac{1}{2}m\left(\frac{1}{4}L^2\dot{\theta}^2\right) + \frac{1}{2}\times\frac{1}{12}mL^2\dot{\theta}^2 + \frac{1}{2}m\left(\frac{1}{4}L^2\dot{\theta}^2 + 2L^2\sin^2\,\theta\cdot\dot{\theta}^2\right) + \frac{1}{2}\times\frac{1}{12}mL^2\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2$$
$$= \frac{1}{6}mL^2\dot{\theta}^2 + \frac{1}{6}mL^2\dot{\theta}^2 + mL^2\sin^2\,\theta\cdot\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2$$
$$= \frac{1}{3}mL^2\dot{\theta}^2 + mL^2\sin^2\,\theta\cdot\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2$$
$$V = -mg\frac{L}{2}\sin\,\theta - mg\frac{L}{2}\sin\,\theta + \frac{1}{2}k_1(2L - 2L\cos\,\theta + x)^2 + \frac{1}{2}k_2x^2$$
$$L = T - V$$
$$= \frac{1}{3}mL^2\dot{\theta}^2 + mL^2\sin^2\,\theta\cdot\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2 + mgL\sin\,\theta - \frac{1}{2}k_1(2L - 2L\cos\,\theta + x)^2 - \frac{1}{2}k_2x^2$$

3. Hamilton's principle,

$$\delta W_{nc} = 0$$

$$\int_{t_1}^{t_2} (\delta L + \delta W_{nc}) dt = 0$$

$$= \int_{t_1}^{t_2} \left(\frac{2}{3} mL^2 \dot{\theta} + 2mL^2 \sin^2 \theta \cdot \dot{\theta}\right) \delta\dot{\theta} + M\dot{x}\delta\dot{x} - (k_1(2L - 2L\cos\theta + x) + k_2 x)\delta x = 0$$

$$+\left(2mL^2 \sin\theta \cos\theta \dot{\theta}^2 + mgL\cos\theta - 2k_1 L\sin\theta(2L - 2L\cos\theta + x)\right)\delta\theta dt$$

$$\downarrow \int_{t_1}^{t_2} f(q,\dot{q})\delta\dot{q} dt = \int_{t_1}^{t_2} f(q,\dot{q}) \frac{d}{dt} \delta q dt = f(q,\dot{q}) \cdot \delta q|_{t_2}^{t_1} - \int_{t_1}^{t_2} \frac{d}{dt}(f(q,\dot{q}))\delta q dt$$

$$\therefore \int_{t_1}^{t_2} + \left\{-\frac{d}{dt}\left(\frac{2}{3} mL^2 \dot{\theta} + 2mL^2 \sin^2 \theta \dot{\theta}\right) + 2mL^2 \sin\theta \cos\theta \dot{\theta}^2 + mgL\cos\theta - 2k_1 L\sin\theta(2L - 2L\cos\theta + x)\right\}\delta\theta$$

$$dt$$

$$\therefore \begin{cases} \left(\frac{2}{3} mL + 2mL^2 \sin^2 \theta\right)\ddot{\theta} + 2mL^2 \sin\theta \cos\theta \dot{\theta}^2 - mgL\cos\theta + 2k_1 L\sin\theta(2L - 2L\cos\theta + x) = 0 \\ M\ddot{x} + k_1(2L - 2L\cos\theta + x) + k_2 x = 0 \end{cases}$$

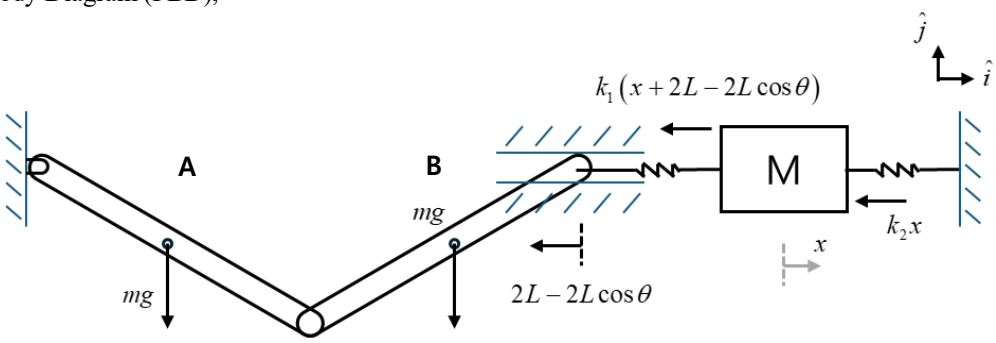
3 (Total 25 pts.) Derive the equilibrium equations by means of <u>the virtual work principle</u> in the system as shown in **Fig. 2.25.** The angle θ can be arbitrarily large. Follow the questions step by step.

1) **(10 pts)** Express all forces acting on the system in terms of unit vectors within the designated coordinate system shown in **Fig. 2.25** and write the corresponding virtual displacements including δθ.

   (**Hint**: $F_i = \mathrm{a}\,\hat{\imath} + \mathrm{b}\,\hat{\jmath},\ \delta x_i = \mathrm{c}\delta x + \mathrm{d}\delta\theta,\ i = 1, 2, 3, \ldots$ and a, b, c, d are arbitrary constants.)

2) **(15 pts)** Based on the forces and virtual displacements obtained in 1), apply the principle of virtual work and derive the equation of motion of the system.

1. Free Body Diagram (FBD),



2. Equations of motion (virtual work principle),

$$F_1 = -k_1(x + 2L - 2L\cos\,\theta)\hat{\imath} \quad F_2 = -k_2x\hat{\imath} \quad F_3 = -mg\hat{\jmath} \quad F_4 = -mg\hat{\jmath}$$

$$\delta x_1 = \delta x + 2L\sin\,\theta\delta\theta \quad \delta x_2 = \delta x_2 \quad \delta x_3 = -\frac{L}{2}\cos\,\theta\delta\theta \quad \delta x_4 = -\frac{L}{2}\cos\,\theta\delta\theta$$

$$\begin{aligned}\sum F_i\delta x_i &= -k_1(x + 2L - 2L\cos\,\theta)\delta x - k_1(x + 2L - 2L\cos\,\theta)2L\sin\,\theta\delta\theta \\ &\quad -k_2x\delta x + \frac{mg}{2}L\cos\,\theta\delta\theta + \frac{mg}{2}L\cos\,\theta\delta\theta \\ &= (-2k_1L(x + 2L - 2L\cos\,\theta)\sin\,\theta + mgL\cos\,\theta)\delta\theta \\ &\quad +(-k_1(x + 2L - 2L\cos\theta) - k_2x)\delta x\end{aligned}$$

$$\therefore \begin{cases} -2k_1L\sin\,\theta(x + 2L - 2L\cos\,\theta) + mgL\cos\,\theta = 0 \\ (k_1 + k_2)x + 2k_1L(1 - \cos\,\theta) = 0 \end{cases}$$

4 (Total 25 pts.) Derive the <u>Lagrange equation</u> of motion for the system as shown in **Fig 2.25.** Follow the questions step by step.

1) **(5 pts)** Identify the number of generalized coordinates required to describe the motion of the system, specify how many there are, and denote them with appropriate symbols.

2) **(20 pts)** Using the Lagrange equation of motion, derive the equation of motion for the system.

(**Hint**: $\frac{\mathrm{d}}{\mathrm{dt}}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0, \quad i = 1, 2, 3, \ldots, \mathrm{N},$

N: The number of generalized coordinates)

1. The number of generalized coordinates required to describe the motion of the system is 2 → (x, θ)
2. Equations of motion,

$$r_A = \frac{L}{2}\cos\,\theta\hat{\imath} - \frac{L}{2}\sin\,\theta\hat{\jmath} \qquad v_A = -\frac{L}{2}\sin\,\theta\cdot\dot{\theta}\hat{\imath} - \frac{L}{2}\cos\,\theta\cdot\dot{\theta}\hat{\jmath}$$
$$r_B = \frac{3}{2}L\cos\,\theta\hat{\imath} - \frac{L}{2}\sin\,\theta\hat{\jmath} \qquad v_B = -\frac{3}{2}L\sin\,\theta\cdot\dot{\theta}\hat{\imath} - \frac{L}{2}\cos\,\theta\cdot\dot{\theta}\hat{\jmath}$$
$$r_C = x\hat{\imath} \qquad v_C = \dot{x}\hat{\imath}$$

3. Lagrangian,

$$T = \frac{1}{2}m\left(\frac{1}{4}L^2\dot{\theta}^2\right) + \frac{1}{2}\times\frac{1}{12}mL^2\dot{\theta}^2 + \frac{1}{2}m\left(\frac{1}{4}L^2\dot{\theta}^2 + 2L^2\sin^2\,\theta\cdot\dot{\theta}^2\right) + \frac{1}{2}\times\frac{1}{12}mL^2\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2$$
$$= \frac{1}{6}mL^2\dot{\theta}^2 + \frac{1}{6}mL^2\dot{\theta}^2 + mL^2\sin^2\,\theta\cdot\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2 = \frac{1}{3}mL^2\dot{\theta}^2 + mL^2\sin^2\,\theta\cdot\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2$$

$$V = -mg\frac{L}{2}\sin\,\theta - mg\frac{L}{2}\sin\,\theta + \frac{1}{2}k_1(2L - 2L\cos\,\theta + x)^2 + \frac{1}{2}k_2x^2$$
$$L = T - V$$
$$= \frac{1}{3}mL^2\dot{\theta}^2 + mL^2\sin^2\,\theta\cdot\dot{\theta}^2 + \frac{1}{2}M\dot{x}^2 + mgL\sin\,\theta - \frac{1}{2}k_1(2L - 2L\cos\,\theta + x)^2 - \frac{1}{2}k_2x^2$$

Lagrange equations, $\begin{cases} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0 \\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0 \end{cases}$

$$\begin{cases} M\ddot{x} - (-k_1(2L - 2L\cos\,\theta + x) - k_2x) = 0 \\ \left(\frac{2}{3}mL^2 + 2mL^2\sin^2\,\theta\right)\ddot{\theta} + 4mL^2\sin\,\theta\cos\,\theta\dot{\theta}^2 \\ -\left(2mL^2\sin\,\theta\cos\,\theta\dot{\theta}^2 + mgL\cos\,\theta - k_1(2L - 2L\cos\,\theta + x)2L\sin\,\theta\right) = 0 \end{cases}$$

$$\therefore \begin{cases} M\ddot{x} + (k_1 + k_2)x + 2k_1L(1 - \cos\,\theta) = 0 \\ \left(\frac{2}{3}mL^2 + 2mL^2\sin^2\,\theta\right)\ddot{\theta} + 2mL^2\sin\,\theta\cos\,\theta\dot{\theta}^2 - mgL\cos\,\theta + 2k_1L(2L - 2L\cos\,\theta + x)\sin\,\theta = 0 \end{cases}$$